Alejandro Buendia

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Research Interests

Machine learning methods for genomic regulation, spatially-resolved data, and the adaptive immune response.

Education

Ph.D. Biomedical Data Science, Stanford University — Sept. 2024 – present
Advisors: Anshul Kundaje and Scott D. Boyd.

B.A. Computer Science and Mathematics, Columbia University — Feb. 2019
Undergraduate research advisor: Daniel Hsu.

Professional Experience

Columbia University Irving Medical Center, New York, NY — Sept. 2023 – July 2024
Computational Research Lab Manager

Massachusetts Institute of Technology, Cambridge, MA — Feb. 2022 – June 2023
Research Engineer / Junior Research Scientist, CSAIL

Microsoft, Cambridge, MA — Jan. 2019 – Jan. 2022
Data Scientist II / Software Engineer

Publications

Journal Articles

Samuel B. Hayward, Alina Vaitsiankova, Tomas Lama-Diaz, Juihsuan Chou, Angelo Taglialatela, Jen-Wei Huang, Yodharaudshani Wijesekarahanthi, Joshua R. Heyza, Giuseppe Leuzzi, Chuanyuan Chen, Nancy Wong, Tenzin Lhakhang, Xi Fu, **Alejandro Buendia**, Veronica Gheorghe, Nazanin Esmaeili Anvar, Jens C. Schmidt, Andre Nussenzweig, Raul Rabadan, Vincenzo Costanzo, Raphael Guerois, Traver Hart, and Alberto Ciccia. Mapping the genetic landscape of the DNA damage response with Cas12a-based combinatorial knockout screens. *Nature Biotechnology* (accepted in principle). 2026

Xi Fu*, Shentong Mo*, **Alejandro Buendia***, Anouchka Laurent, Anqi Shao, Maria del Mar Alvarez-Torres, Tianji Yu, Jimin Tan, Jiayu Su, Romella Sagatelian, Adolfo A. Ferrando, Alberto Ciccia, Yanyan Lan, David M. Owens, Teresa Palomero, Eric P. Xing, and Raul Rabadan. A foundation model of transcription across human cell types. *Nature*. 2025

Rebecca Boiarsky, Nalini M. Singh, **Alejandro Buendia**, Ava P. Amini, Gad Getz, and David Sontag. Deeper evaluation of a single-cell foundation model. *Nature Machine Intelligence*. 2024

Refereed Conference and Workshop Proceedings

Alejandro Buendia*, Chiho Im*, Anshul Kundaje, Scott D. Boyd, Liel Cohen-Lavi. BenchRep-T: a systematic evaluation of T-cell repertoire-based disease diagnostics. *NeurIPS 2026 Evaluations & Datasets*. 2026

Rebecca Boiarsky, Nalini Singh, **Alejandro Buendia**, Gad Getz, and David Sontag. A deep dive into single-cell RNA sequencing foundation models. In *Machine Learning in Computational Biology (MLCB)*. Oral presentation (top 15%). 2023

Stefan Hegselmann, **Alejandro Buendia**, Hunter Lang, Monica Agrawal, Xiaoyi Jiang, and David Sontag. TabLLM: few-shot classification of tabular data with large language models. In *Twenty-Sixth International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2023

Liqun Shao, Sahitya Mantravadi, Tom Manzini, **Alejandro Buendia**, Manon Knoertzer, Soundar Srinivasan, and Chris Quirk. Examination and extension of strategies for improving personalized language modeling via interpolation. In *First Workshop on Natural Language Interfaces (ACL)*. 2020

Daniel Boley and **Alejandro Buendia**. Random walk fundamental tensor and graph importance measures. In *International Workshop on Big Social Media Data Management and Analysis (IJCAI)*. 2019

Alejandro Buendia and Daniel Boley. Optimized graph-based trust mechanisms using hitting times. In *International Workshop on Trust in Agent Societies (AAMAS)*. 2017

Preprints and Manuscripts Under Review

Anvita Gupta, **Alejandro Buendia**, Anshul Kundaje, and Jure Leskovec. Bio-BLIP: a multimodal architecture for transferable reasoning in genomic variant interpretation. *bioRxiv*. 2026

Eric D. Sun*, **Alejandro Buendia***, Anne Brunet, and James Zou. SpatialProp: tissue perturbation modeling with spatially resolved single-cell transcriptomics. *bioRxiv*. 2025

Teaching and Mentoring

Graduate Teaching Assistant, Foundation Models for Healthcare, Stanford University.

Undergraduate Teaching Assistant, Calculus IV and Discrete Mathematics, Columbia University.

Mentored one master's thesis and two undergraduate researchers in clinical natural language processing and product deployment, MIT Clinical Machine Learning Group.

Service and Outreach

Editorial Board Member (current), *Genomics x AI Blog*. A cross-institution, open-source initiative publishing research findings and software for the genomics and AI communities.

Lead (current), qBMI (LGBTQ+ affinity group), Department of Biomedical Data Science, Stanford University.

Co-founder and lead, Diversity and Inclusion Initiative, Microsoft AI Development Program. Recruited underrepresented-minority graduates at academic and job-search conferences; partnered with Black in AI, LatinX in AI, and Women in ML.

Microsoft TEALS Outreach Program. Led a team of six developing a three-module introductory AI course for high school students; presented career talk about working at Microsoft to high school students in the Boston area.

Writing

Benchmarking AlphaGenome on NVIDIA GPUs: latency, memory, and feasibility across sequence lengths. *Genomics x AI Blog.* 2026

Porting AlphaGenome to PyTorch. *Genomics x AI Blog.* 2026

Fine-tuning AlphaGenome in native JAX/Haiku. *Genomics x AI Blog.* 2026

Software

`alphagenome_ft`: Fine-tuning utilities for AlphaGenome. Core developer.

`alphagenome-pytorch`: PyTorch implementation of AlphaGenome. Core developer.

Awards and Fellowships

Warren Alpert Computational Biology & Artificial Intelligence Fellowship, Stanford University.

MAC3 Computing in Precision Health Fellowship, Stanford University.

Kluge Scholar, Columbia University.

Columbia College Summer Funding Program, Columbia University.